Mains input & heater

File: mains_heater.kicad_sch

Motor DC bus (doubler) & pre−charge

File: motor_bus.kicad_sch

Motor power stage (4−phase SR)

File: motor_stage.kicad_sch

Safety logic (relay permits, watchdog, gate−drive enable)

File: safety_logic.kicad_sch

Analog sensing (currents, trip, motor sensors)

File: analog_sense.kicad_sch

MCU, programming header & front−panel link

File: mcu_links.kicad_sch

Low−voltage power supply (flyback)

File: power_supply.kicad_sch

THERMOMIX TM31−4C (120 V) POWER BOARD NPVTM31LTA−3 − REVERSE−ENGINEERED SCHEMATIC rev 0.2 (2026−11)

HOW TO READ THIS PROJECT
− Each sheet is drawn as a real circuit: parts are wired together; supply rails use power symbols; only signals that leave a sheet use
  global labels (flag−shaped). A few busy nodes inside a sheet use local net labels (plain text on a wire); same name = same net.
− Relays are split into two units: contacts (A) where they switch power, coil (B) next to its driver on 'Safety logic'.
  The dual op−amps are split into halves A/B and a supply unit C; the watchdog 74HC123 into timers A/B and supply C.
− References are the board's silkscreen references. RXn / CXn / DXn = part present on the board whose reference was not read.
− '?' = value not read. 'approx' / APPROXIMATE = inferred from photos or partial tracing, not confirmed by continuity.
− Everything except the CN300 'COLD' corner is at mains potential. Off−board items (bowl heater, motor windings, motor NTC) are drawn
  in dashed frames and excluded from the BOM.
− The connections are identical to rev 0.1 (verified by comparing the exported netlists node by node).

ARCHITECTURE IN ONE PARAGRAPH
Mains −> EMI filter −> two relays (RY200 live, RY202 neutral) −> heater (through a triac and current transformer T700) and the motor DC bus
(a voltage doubler giving ~340 V DC, pre−charged through R115/R114 and then bypassed by RY201). A 78K0 MCU on this board drives a 4−phase
switched−reluctance motor (6 IGBTs, shared high−side switches) and talks to the front panel over an opto−isolated UART (CN300).
Hardware safety: both mains relays and the IGBT gate−drive supply need (a) two open−drain MCU permits that only IC700's fault comparators can hold
high, (b) a running 74HC123 watchdog fed by MCU pulses, and (c) for RY202, the front panel's enable through PC503. A comparator (IC610) cuts the
high−side IGBTs in hardware on motor overcurrent.

KNOWN FAILURES / NOTES FROM THIS BOARD
− No heat, no error: CR700 triac A1 leg fatigued and cracked at its C−bend (fixed 2026−10, BTA24−600BW, re−form the C−bend). Remember the TM31 only
  heats while a TIME is set and the blade is turning.
− Shorted triac (230 V forum reports) −> continuous heating, error 69. F100 open / R115 open −> no motor.

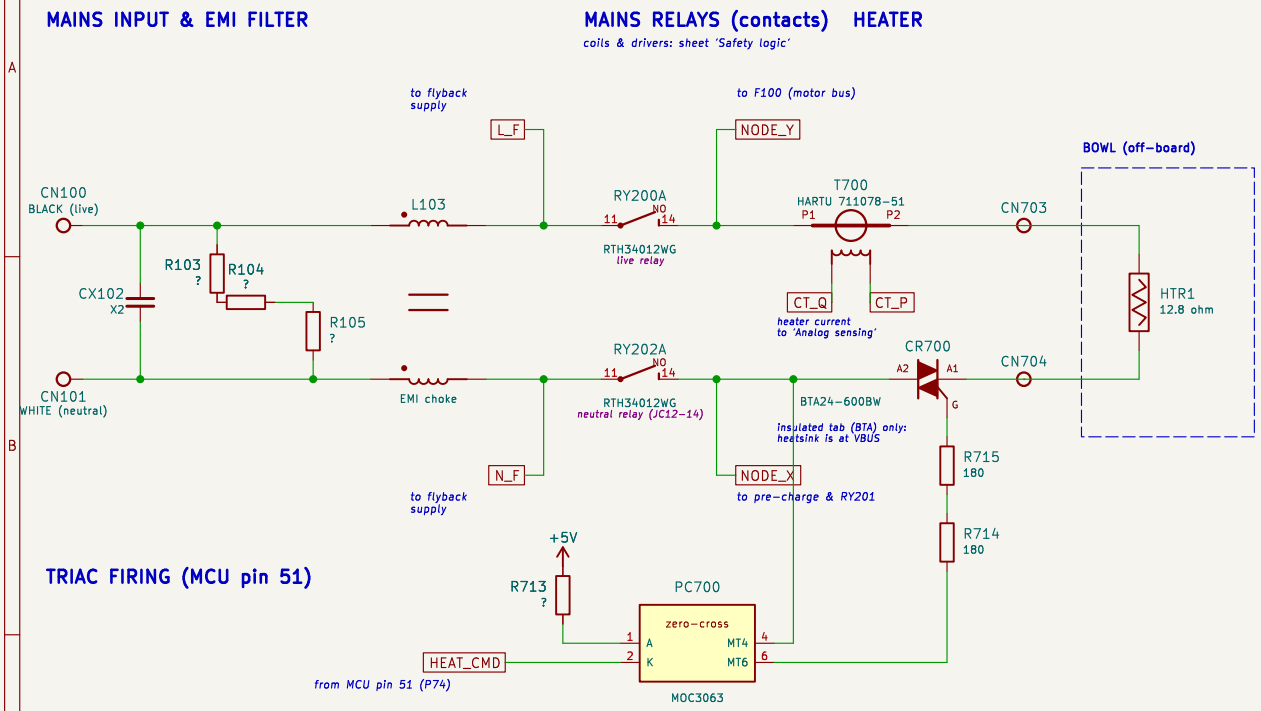
KEY NETS
Power symbols (rails): GND = hot logic ground = motor bus negative (NOT neutral, NOT earth);
  +5V logic; V12RAW relay coils / gate−drive source; VGATE switched gate−drive supply; VNEG negative rail (IC611);
  VBUS ~340 V DC (= heatsink); HVDC / HVDC_N flyback primary; VCOLD / GND_COLD isolated front−panel supply.
Global labels (signals between sheets):
  L_F / N_F ..... live / neutral after the EMI choke L103
  NODE_Y / NODE_X .. live after RY200 / neutral after RY202
  CT_P / CT_Q ..... T700 secondary (heater current)
  LS_EMIT ..... low−side IGBT emitters (top of the shunts)
  PERMIT1/2, WD2 ... relay permits (MCU 11/12), watchdog 2 (MCU 13)
  ARM / WDCLK ..... watchdog arm (panel + MCU 14) / pulses (MCU 52)
  TRIP ..... overcurrent trip (IC610 −> D512/D513, R535 −> MCU 45)
  other labels ..... MCU pin signals, named after the port or function

Thermomix TM31−4C (120 V) power board NPVTM31LTA−3

Sheet: /
File: TM31_power_board.kicad_sch

Title: TM31 power board − overview

Size: A3	Date: 2026−11−03	**Rev: 0.2**
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4		Id: 1/8



NOTES (confirmed by continuity unless marked; ? = value not read)
- CN100 = BLACK tab (live), CN101 = WHITE tab (neutral). CY103/CY104 Y-capacitors exist; their far ends were not traced (not drawn).
- RY200 switches live (L_F) to NODE_Y; RY202 switches neutral (N_F) to NODE_X through three parallel jumpers JC12–JC14.
  Both relays only close when the safety logic permits it.
- Heater path: L_F – RY200 – NODE_Y – T700 – CN703 – heater – CN704 – CR700 A1..A2 – NODE_X – RY202 – N_F. No fuse in the heater path.
- CR700 original marking unreadable (date code 409); replaced with BTA24-600BW after the A1 leg cracked in its C-bend (2026-10).
  Re-form the C-bend on replacement legs. T700 secondary: S1/S2 order assumed.
- The TM31 only heats while a time is set and the blade turns. 120 V bowl: heater 12.8 ohm, NTC ~130 kohm at room temperature.

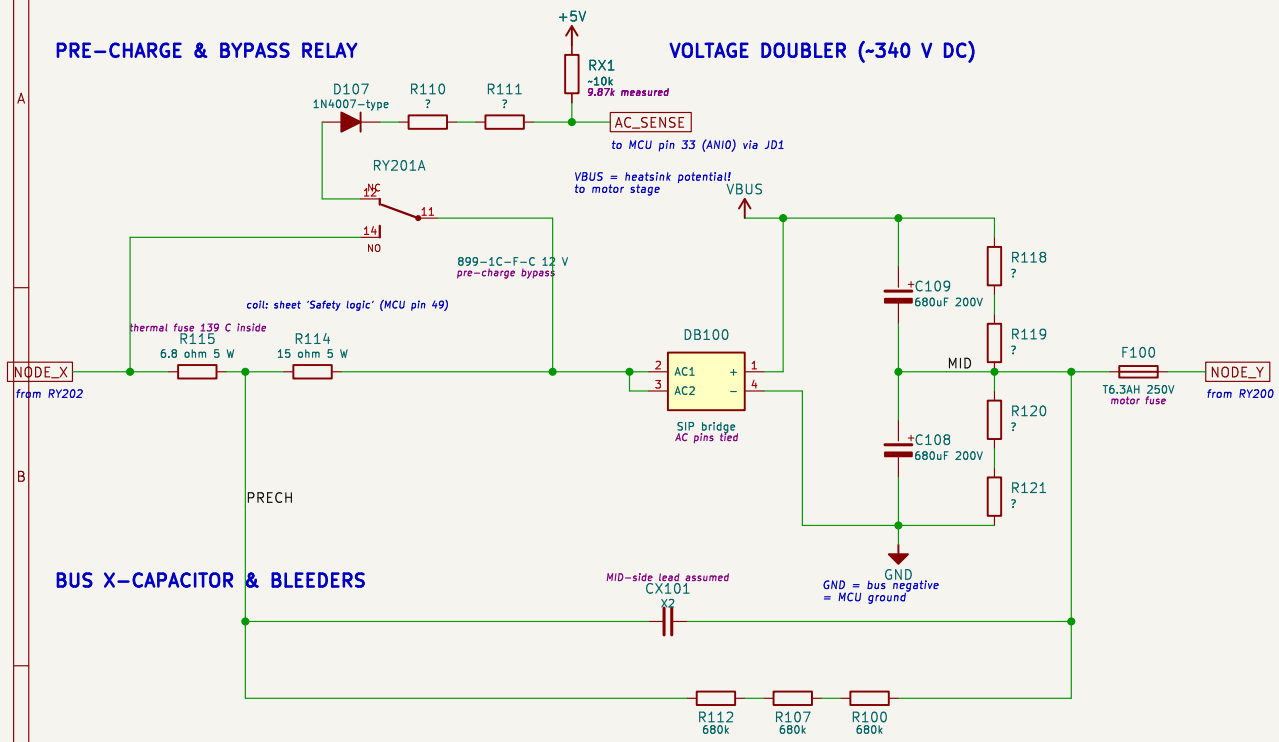
Traced from a working board by continuity tests; see notes for confidence levels

Thermomix TM31-4C (120 V) power board NPVTM31LTA-3

Sheet: /Mains input & heater/
File: mains_heater.kicad_sch

Title: Mains input & heater

Size: A4	Date: 2026-11-03	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/8



NOTES (? = value not read)
– 120 V version of a 230 V design: NODE_Y (live after RY200) feeds the capacitor midpoint MID through F100, and NODE_X (neutral after RY202) reaches BOTH AC pins of DB100 through R115 + R114. Result: VBUS ~340 V DC, the same bus as a 230 V board's bridge.
– Power–up (3 clicks): RY202 + RY200 close, C108/C109 charge through R115 + R114; RY201 then shorts them (COM to NO). At rest, RY201's NC contact routes DB100's AC node to D107 –> R110/R111 –> MCU pin 33, so the MCU can check the pre–charge voltage first.
– F100 open or R115 open: motor dead, heater still works. Measure from GND (bus –) with a floating meter only.
– APPROXIMATE: pairing of balancing resistors R118–R121; CX101's MID–side lead. Not drawn: Z100 (unmarked, probably a varistor), R214.

Traced from a working board by continuity tests; see notes for confidence levels

Thermomix TM31–4C (120 V) power board NPVTM31LTA–3

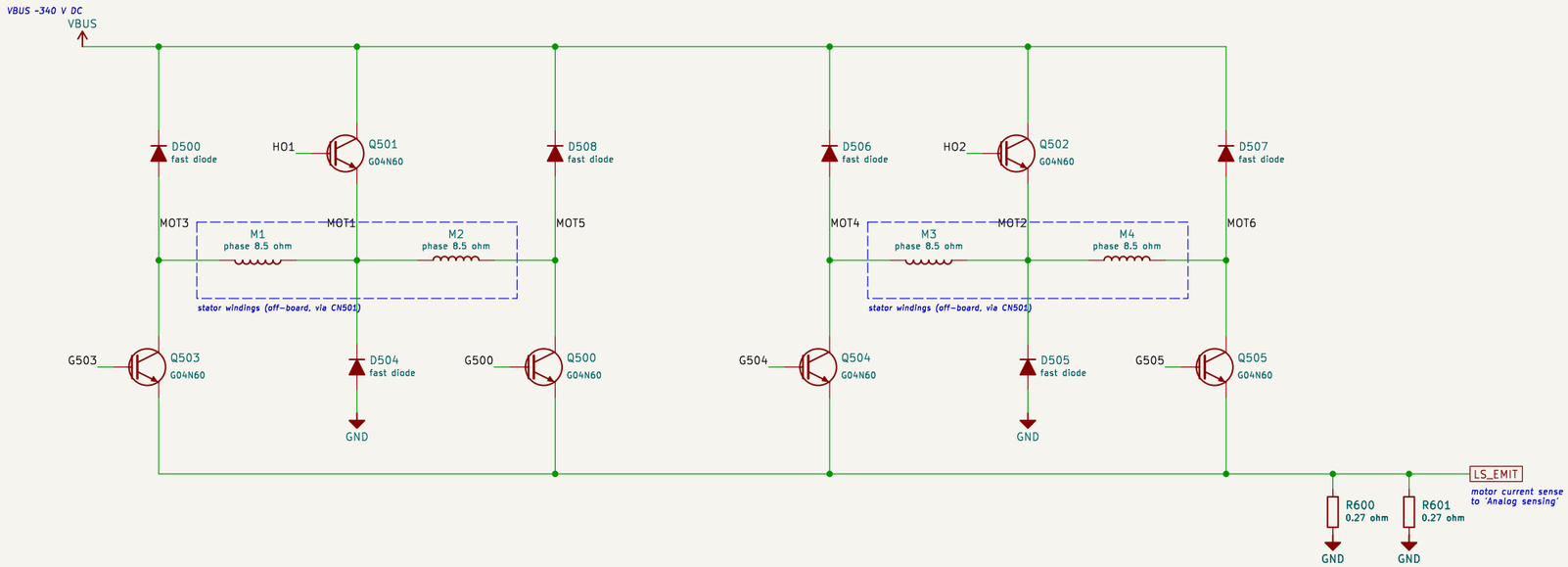
Sheet: /Motor DC bus (doubler) & pre–charge/
File: motor_bus.kicad_sch

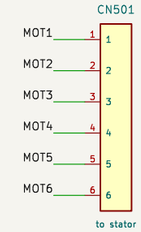
Title: Motor DC bus (doubler) & pre–charge

Size: A4	Date: 2026–11–03	**Rev: 0.2**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 3/8

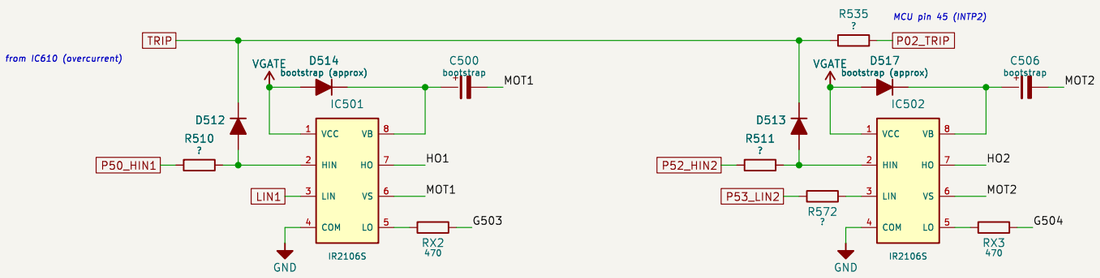
POWER STAGE: 4-PHASE SWITCHED-RELUCTANCE MOTOR, SHARED HIGH-SIDE SWITCHES

phases (stator, 8.5 ohm each): CN501 1-3 and 1-5 share high side Q501; 2-4 and 2-6 share high side Q502



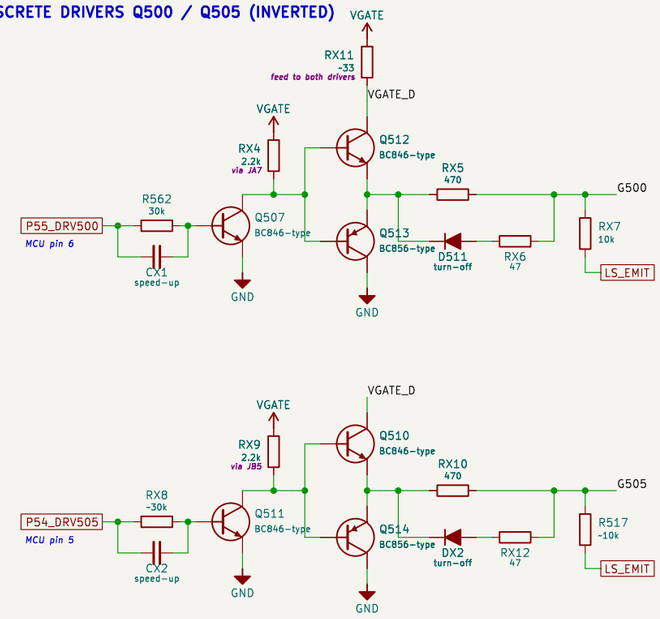


HIGH/LOW-SIDE DRIVERS (Q501-Q504) & HARDWARE OVERCURRENT CUT-OFF



MCU: pin 1 HIN1, 2 LIN1, 3 HIN2, 4 LIN2

DISCRETE DRIVERS Q500 / Q505 (INVERTED)



NOTES (? = value not read)
- IGBTs Q500-Q505: G04N60 (Infineon SGP04N60-type, 600 V). Low-side emitters (LS_EMIT) return to GND through R600 // R601 (0.27 ohm each).
- Freewheel diodes all ~0.475 V in diode mode. IR2106 supply VGATE exists only while the safety logic enables Q205/Q206: a firmware crash or a refused permit removes ALL gate drive. Discrete drivers: MCU pin high = IGBT off: turn-on through 470 ohm, fast turn-off through 47 ohm + diode.
- Hardware overcurrent cut-off: IC610 pulls TRIP low; D512/D513 force both HIN inputs low (high sides off without firmware) and interrupt the MCU (pin 45).
- APPROXIMATE: bootstrap diodes (D514/D517 drawn; D518/D519 also exist); HO gate resistors and Q503/Q504 gate pull-downs not traced; the ~33 ohm feed RX11 may be two resistors; DX2 = turn-off diode of Q505 (reference not read).

Traced from a working board by continuity tests; see notes for confidence levels

Thermomix TM31-4C (120 V) power board NPVTM31LTA-3

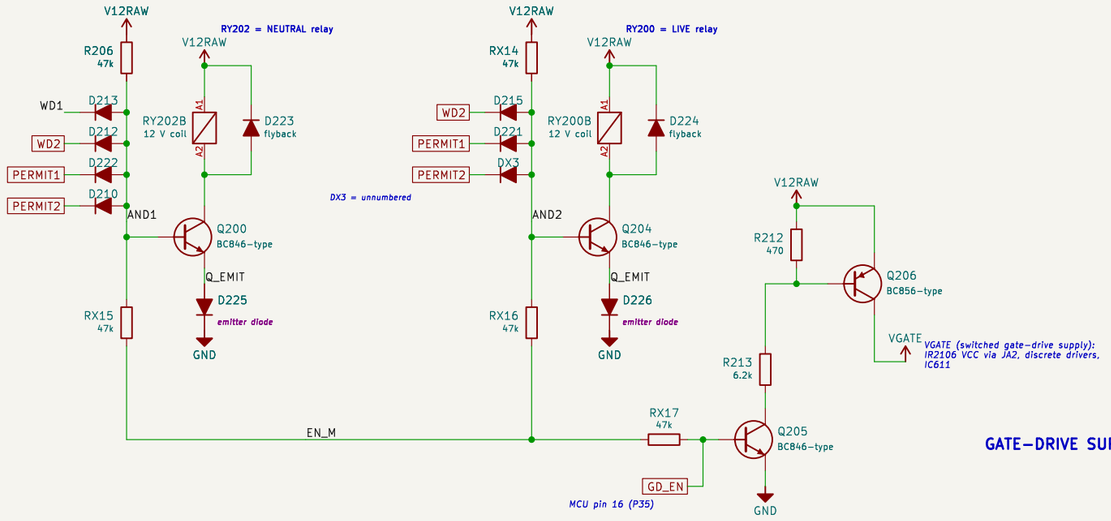
Sheet: /Motor power stage (4-phase SR)/
File: motor_stage.kicad_sch

Title: Motor power stage (4-phase SR)

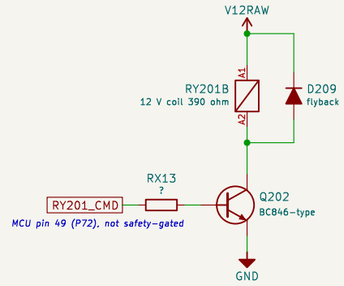
Size: A3	Date: 2026-11-03	**Rev: 0.2**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 4/8

MAINS RELAY PERMITS (diode-AND gates)

a relay closes only if EVERY input is high; any input low drops the AND node

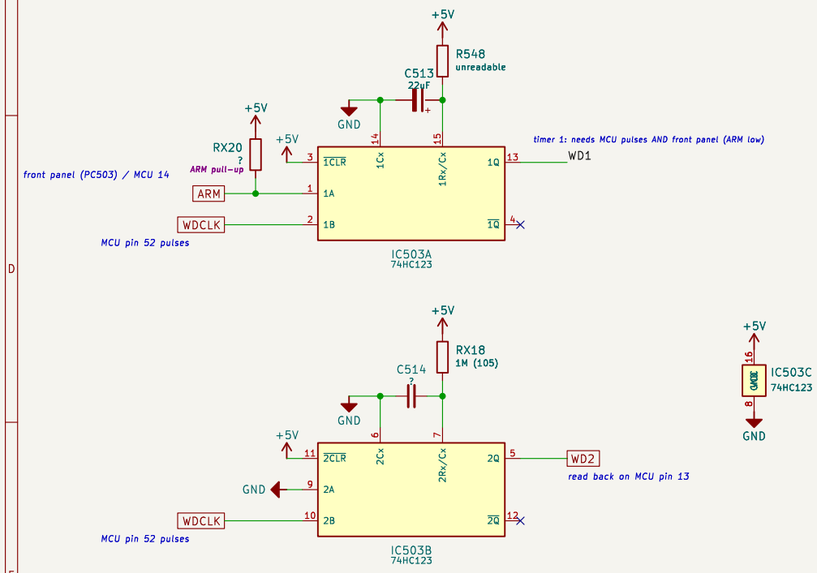


PRE-CHARGE BYPASS RELAY

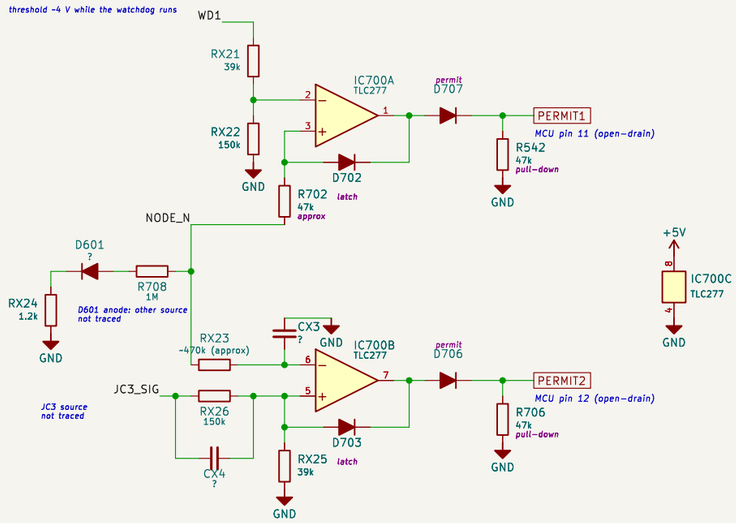


GATE-DRIVE SUPPLY ENABLE

HARDWARE WATCHDOG (IC503 74HC123, dual retriggerable one-shot)



LATCHING FAULT COMPARATORS (IC700 TLC277): the only source that can hold the permits high



NOTES (structure confirmed by continuity; MCU port types from the uPD78F0034 datasheet p.11: ? = value not read)
- MCU pins 11-14 (P30-P33) are N-channel OPEN-DRAIN: they can only pull a line low, or read it. PERMIT1/2 are pulled DOWN by R542/R706, so only IC700 (through D707/D706) can hold them high. A dead or disconnected detector therefore removes the permit (fail-safe). The MCU can veto (pull low) and read them.
- VGATE: the 3 x 47 kohm network (RX15, RX16, RX17) feeds Q205 from BOTH AND nodes, so the IGBT drivers are powered only while both mains relays are permitted.
- Watchdog: timer 1 time-out ~0.45 x R548 x 22 uF, timer 2 ~0.45 x 1 Mohm x C514. Q_EMIT: D225/D226 raise the relay drivers' turn-on threshold; series/parallel not certain.
- IC700: D702/D703 feed each output back to its own + input (latch). Half A's threshold comes from WD1, so the latch clears when the watchdog restarts.
- APPROXIMATE: the sensed signals (via D601/R708 and the JC3 path) were only partly traced. Most likely role: heater current while the triac should be off (error 69).

Traced from a working board by continuity tests; see notes for confidence levels

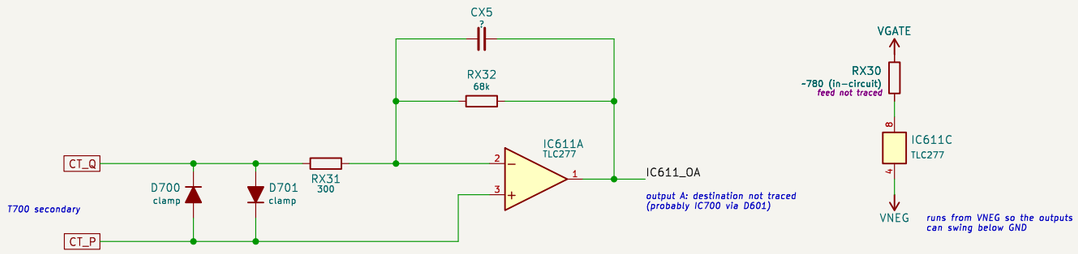
Thermomix TM31-4C (120 V) power board NPVTM31LTA-3

Sheet: /Safety logic (relay permits, watchdog, gate-drive enable)/
File: safety_logic.kicad_sch

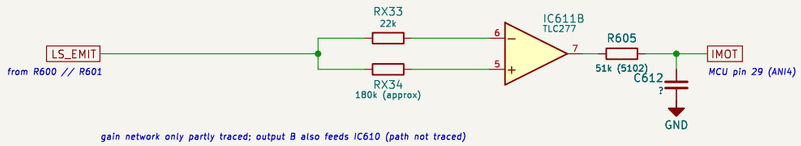
Title: Safety logic (relay permits, watchdog, gate-drive enable)

Size: A3	Date: 2026-11-03	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/8

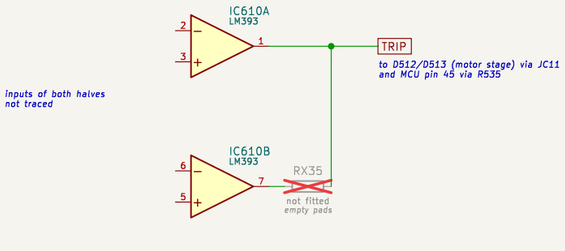
HEATER CURRENT AMPLIFIER (IC611 A, from T700)

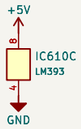


MOTOR CURRENT AMPLIFIER (IC611 B, from the shunts)

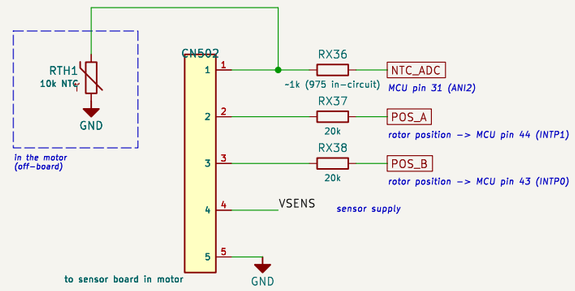


OVERCURRENT COMPARATOR (IC610 LM393)

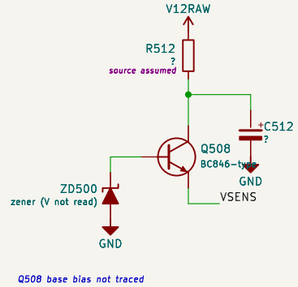




MOTOR SENSOR BOARD (CN502)



SENSOR SUPPLY (zener follower)



NOTES (approximate where marked; ? = value not read)
- IC611 TLC277 is powered between pin 8 (~0.78 kohm in-circuit to VGATE; exact feed not traced) and the negative rail VNEG (pin 4, C230).
- Half A converts the T700 secondary current to a voltage (300 ohm in, 68 kohm // C feedback, D700/D701 clamp). Where CT_P is referenced and where output A goes were not traced.
- Half B amplifies the shunt voltage (LS_EMIT) for the MCU (pin 29). IC610 output A is the hardware overcurrent trip; output B goes to unpopulated pads (second trip, not fitted).
- CN502: pin 1 motor NTC (~12.5 kohm at room temperature, i.e. a 10 kohm NTC); pins 2/3 rotor-position sensors; pin 4 sensor supply; pin 5 GND.

Traced from a working board by continuity tests; see notes for confidence levels

Thermomix TM31-4C (120 V) power board NPVTM31LTA-3

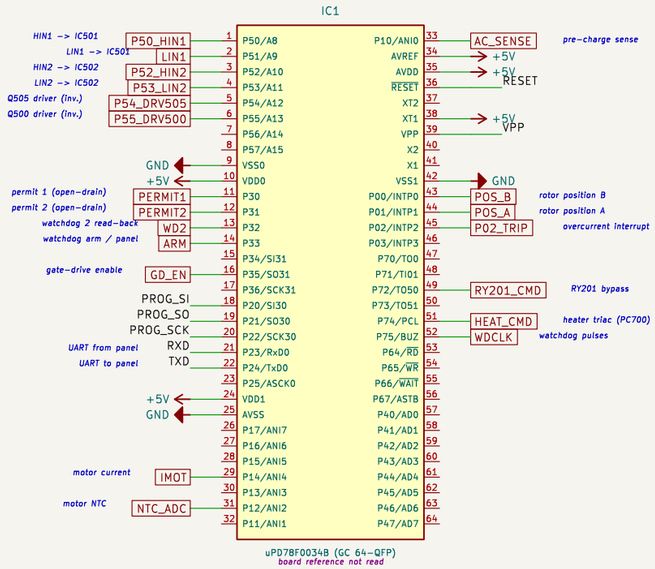
Sheet: /Analog sensing (currents, trip, motor sensors)/
File: analog_sense.kicad_sch

Title: Analog sensing (currents, trip, motor sensors)

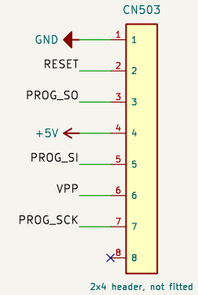
Size: A3	Date: 2026-11-03	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/8

MICROCONTROLLER NEC/Renesas uPD78F0034B (78K0, 64-pin QFP)

untraced pins left open; crystal/resonator (X1/X2) and reset circuit not traced

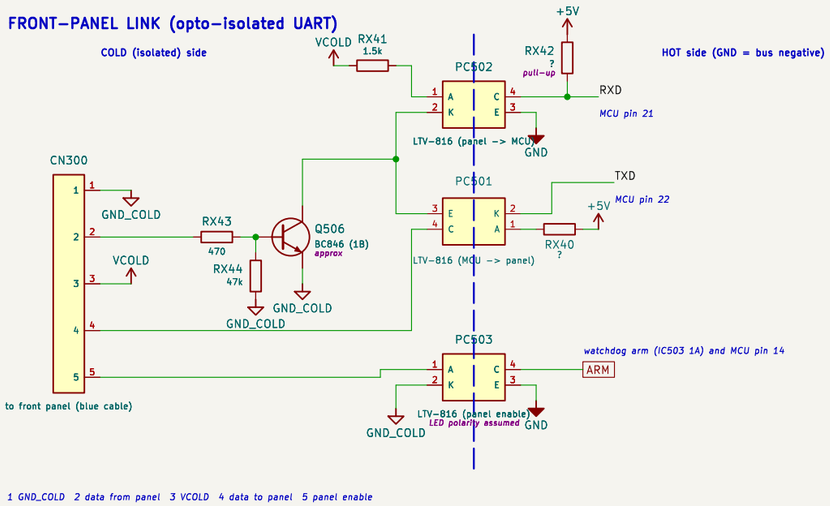


FLASH PROGRAMMING HEADER CN503



3-wire CSI30 flash mode;
not connected in the machine

FRONT-PANEL LINK (opto-isolated UART)



NOTES
- MCU pin numbers per the datasheet QFP drawing (p.6), marking D78F0034BCC (archived datasheet in datasheets/archive/).
- UART0: PC502 carries panel -> MCU RxD0 (pin 21); PC501 carries MCU TxD0 (pin 22) -> panel. PC503 (LED driven by the panel on pins 5/1) pulls ARM low = arms watchdog timer 1.
- APPROXIMATE on the cold side: Q506's exact role, LED polarities of PC501/PC503, resistor values RX40-RX42; R540 not drawn.
- Never connect a mains-earthed adapter to the hot side. To sniff the link, use the COLD side of CN300 only (see docs/15-future-work.md).

Traced from a working board by continuity tests; see notes for confidence levels

Thermomix TM31-4C (120 V) power board NPVTM31LTA-3

Sheet: /MCU, programming header & front-panel link/
File: mcu_links.kicad_sch

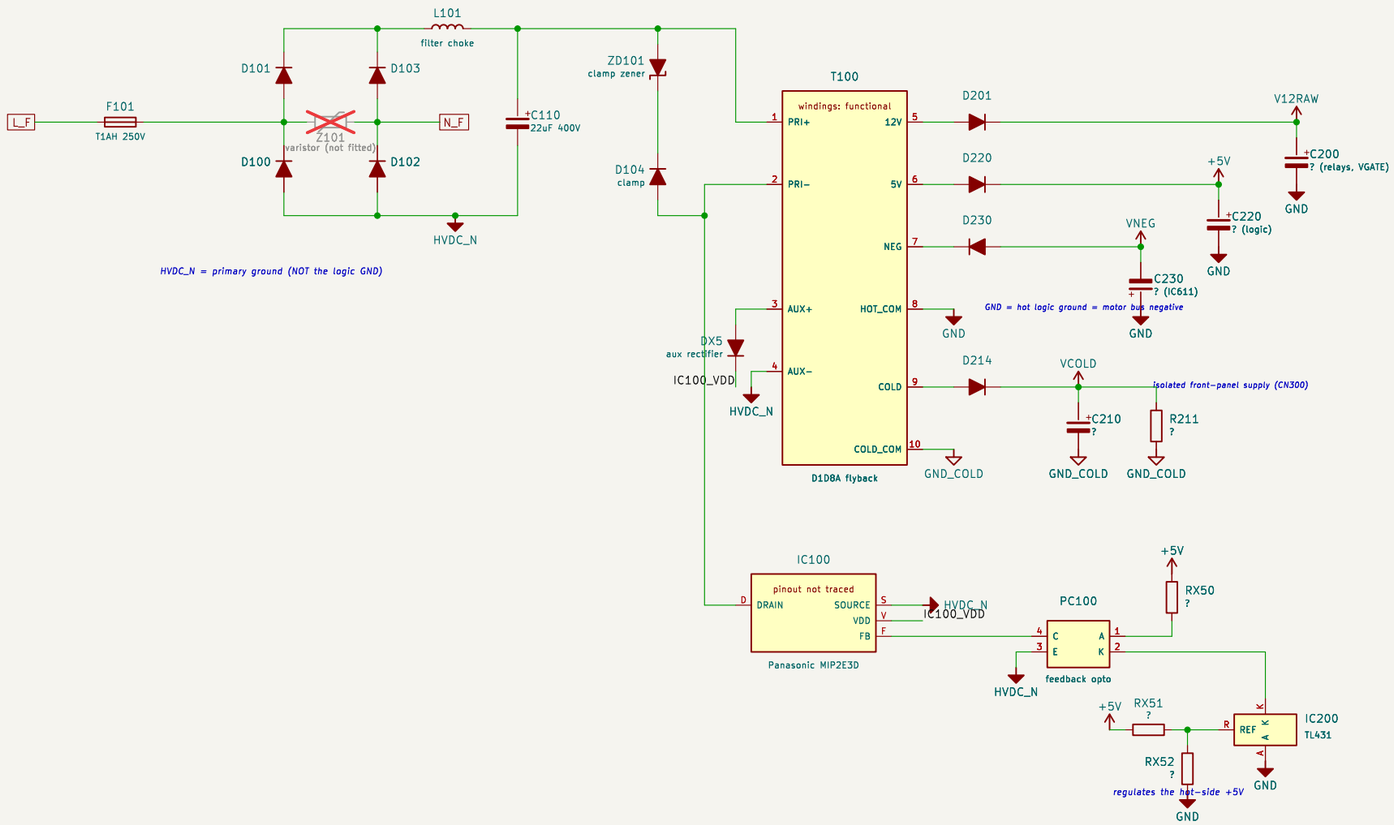
Title: MCU, programming header & front-panel link

Size: A3	Date: 2026-11-03	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/8

INPUT RECTIFIER & FILTER (fed before the relays)

FLYBACK CONVERTER

SECONDARY RAILS



NOTES (structure confirmed; transformer pin-out and several values NOT traced; ? = value not read)
- Fed from mains BEFORE the relays (it must power the MCU before RY200/RY202 close). Exact tap (before/after L103) not confirmed; drawn after L103.
- F101 T1 A; Z101 varistor position empty on this board; IC100 Panasonic MIP2E3D (forum reports MIP2E1D on 230 V boards); T100 marked D1D8A.
- Feedback: PC100 + IC200 TL431 regulate the hot-side +5V. Clamp ZD101/D104, the aux winding and the IC100 pin-out are drawn functionally (DX5 = aux rectifier).
- Hot-side outputs are referenced to GND (= motor bus negative = MCU ground). The cold output (VCOLD / GND_COLD) is the only isolated rail.

Traced from a working board by continuity tests; see notes for confidence levels

Thermomix TM31-4C (120 V) power board NPVTM31LTA-3

Sheet: /Low-voltage power supply (flyback)/
File: power_supply.kicad_sch

Title: Low-voltage power supply (flyback)

Size: A3	Date: 2026-11-03	**Rev: 0.2**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/8